

## Contact

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## Skills

### Programming & Software

- Python
- C/C++
- MATLAB/Simulink
- Git
- Synopsys
- Linux
- Shell/Bash
- Jinja
- ELDO
- Virtuoso

### Characterization & Hardware

- Electrical characterization
- Probe-card testing
- STM32 / FreeRTOS
- Vivado (FPGA)
- Microchip Studio
- Arduino IDE

### Languages

- Portuguese (native language)
- French (fluent)
- English (advanced) – 900 TOEIC (2024)

## Publications

- Selective Algorithm for Expanded Group Method of Data Handling Applied to Power Amplifier Modeling*. JICS, v. 17, p. 1-8, 2022.
- Selective Method with Adapted Activation Function and Applied in Non-linear Optimization for Modeling of Power Amplifiers*. In: 38th South Symposium on Microelectronics, 2023, Porto Alegre.
- Real-valued Neural Network Based in Group Method of Data Handling Applied to Power Amplifier Modeling*. In: XXII Microelectronics Students Forum, 2022, Minuano.
- Selective Algorithm with Different Extraction Methods for Power Amplifier Modeling Based on GMDH*. In: Paraná Microelectronics Seminars, 2022, Curitiba.
- Selective Algorithm for Group Method of Data Handling Applied to Power Amplifier Modeling*. In: XXI Microelectronics Students Forum, 2021, Campinas.

→ Research activities, publications, and academic background available on the [Lattes](#) Platform.

## Scholarships

BRAFITEC Scholarship – French-Brazilian Engineering Exchange Program (2023–2025).

# Ana Paula PRINCIVAL MACHADO

PhD Candidate in Electronics at CEA-Leti and Université Grenoble Alpes, working on embedded AI, emerging memories, and neuromorphic computing.

## Experience

### October 2025 – Present | CEA-Leti | France CEA-Leti – Grenoble

#### PhD Candidate in Electronics

- Electrical characterization of FeFET memory devices using probe card testing.
- Implementation of electrical characterization and reliability test protocols.
- Data processing, analysis, and visualization using Python.
- SPICE simulation of ferroelectric memory devices.
- Research on emerging memory technologies and in-memory computing architectures.

### February 2025 – August 2025 | NXP Semiconductors | France NXP Semiconductors – Toulouse

#### Behavioral Modeling Platform Intern

- Python programming for automating model generation.
- Use of Linux, Git, Jira and Jinja in a technical environment.
- Contribution to a modeling platform for AMS circuits.
- Support for the integration of simulation tools, including EDA and Verilog-AMS.

### September 2024 – February 2025 | ENSICAEN | France

#### D\_RACE Immersive Project | NXP CUP Industrial Project – Caen Instrumentation Systems Developer

- Immersive Project: Development of a remote-controlled vehicle based on STM32.
- Responsible for the integration of electronic components (LEDs, sensors).
- NXP CUP Project: Conversion of the vehicle into an autonomous line-following car.
- Responsible for training a YOLO neural network for track detection.

### April 2024 – August 2024 | GANIL | France

#### Grand Accélérateur National d'Ions Lourds (GANIL) – Caen Software Development Intern – Data Acquisition Group

- Developed an endianness conversion method for a data acquisition library.
- Implemented the conversion for 32 acquisition data frame formats.
- Developed software in C++ using Linux, CentOS and Bash.

### September 2020 – August 2023 | GICS | Brazil

#### Group of Integrated Circuits and Systems – Curitiba Undergraduate Researcher

- Developed neural network models for nonlinear power amplifier behavioral modeling.
- Designed and trained GMDH-based neural networks using MATLAB.
- Optimized model performance (MSE minimization and network parameter tuning).
- Achieved a modeling accuracy of –40 dB NMSE, resulting in multiple scientific publications.

## Academic Background

### August 2023 – September 2025 | ENSICAEN | France

#### Engineering Degree (Master's level) in Embedded Systems (SATE)

- Programming, software engineering, and operating systems.
- Computer architecture, networks, and real-time systems.
- Electronic hardware design, PCB routing, and soldering.
- Digital, predictive, robust, and adaptive control systems.
- Telecommunications: modulation, transmission, spectral analysis, and filtering.

### March 2020 – August 2025 | UFPR | Brazil

#### B.Sc. in Electrical Engineering

- Energy conversion, electromagnetics, and power electronics.
- Residential electrical installation design and power systems.
- Analog electronics: amplifiers, analog filters, and frequency response.
- Electronic simulation projects (e.g., 4-bit analog-to-digital converter).

## References

- Elisa Vianello (CEA) – PhD Supervisor – [Elisa.VIANELLO@cea.fr](mailto:Elisa.VIANELLO@cea.fr)
- Miloud Frikel (ENSICAEN) – SATE Director – [miloud.frikel@ensicaen.fr](mailto:miloud.frikel@ensicaen.fr)
- Eduardo Lima (UFPR) – EE Coordinator – [eduardo.lima@ufpr.br](mailto:eduardo.lima@ufpr.br)
- André Mariano (UFPR) – GICS Coordinator – [mariano@ufpr.br](mailto:mariano@ufpr.br)